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On the ratio of rock fracture parameters in static and dynamic conditions

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Abstract

The methods and results of experiments on contact fracture and splitting of sandstone plates by penetration of a hemispherical indenter under quasi-static and dynamic loading conditions are presented. It was found that, in contact fracture of the rock under quasi-static conditions, the work and the specific energy consumption of the splitting process are substantially lower than under impact. It turned out that the forces required to split sandstone plates on impact are approximately twice as high as the forces under static loading conditions.

Keywords: *sandstone plates, hemispherical indenter, quasi-static loading, impact action, contact fracture, splitting, stiffness coefficient.*

Introduction

The study of rock fracture processes by impact and under quasi-static conditions, i.e. with comparatively slow penetration of a rock-breaking tool, has for many years been the subject of numerous scientific investigations. At the same time, there is still no consensus on the similarities and differences in the mechanisms by which these processes proceed.

Thus, paper [1] notes the similarity of the mechanism of layer-by-layer rock fracture under dynamic and static loading, which consists in the fact that at the initial stage of the fracture process under dynamic conditions the surface layer of the rock being broken is dispersed, and on further penetration of the tool the rock undergoes contact hardening, which from the physical point of view appears analogous to quasi-static pressing of a rigid non-deformable punch (die) into the surface of a solid.

The authors of [2] conclude that the final stage of the rock fracture process under both static and dynamic loading — loss of continuity — is only the concluding act, while the fracture process itself is characterized by the successive nucleation and development of structural defects at various scale levels. Moreover, under dynamic loading, as under static loading, the fracture process does not occur instantaneously but is characterized by the accumulation of damage distributed in time, i.e. dynamic fracture is a multi-stage process. In turn, under static loading, microfracture of rocks occurs in a quasi-static mode, while the subsequent development of the main crack occurs in a dynamic mode at a velocity close to the speed of sound in the medium.

Based on the results of work carried out at the Leningrad Mining Institute, the author of [3] notes that, under both dynamic and static pressing of a punch, a compacted core forms in the rock beneath the contact area. As a result of its expansion in a direction approximately perpendicular to

the action of the pressing force, cracks arise that lead to the appearance of spalls on the surface of the blocks or to their rupture. The same study found a substantial increase in the energy required to fracture objects under dynamic conditions compared with static ones. Thus, for fracturing coal along the bedding ($\sigma_{сж}$ = 30 MPa) using a punch with a spherical working surface, these energies are 12.5 and 2.0 J, respectively, which the author explains by an increase in the resistance of rocks to plastic deformation and fracture on impact.

The authors of other studies, for example [4, 5], arrive at the same conclusion. Thus, paper [4] shows that on passing from static chipping with large chips to impact chipping, the energy expended on fracturing anthracite increases from 1.18 to 1.90 MJ/m^3, and that for clay shale from 1.83 to 6.48 MJ/m^3, i.e. by factors of 1.6 and 3.5, respectively.

The authors of [6–8] explain this difference in the specific energy consumption of rock fracture by an increase in the share of energy spent on forming new surfaces in dynamic fracture, since the greatest amount of it is spent on creating the compacted core and only a very small amount on separating pieces from the massif.

While practically all studies agree that the energy consumption of the fracture process under dynamic loading substantially exceeds that under static loading, opinions in the literature differ concerning the difference in the forces required to fracture rock under static and dynamic conditions.

Thus, paper [9] presents data on the impact penetration of a bit tooth into Indiana limestone ($\sigma_{сж}$ = 40–130 MPa) in comparison with the results of its indentation under static conditions. It was found that the deformation diagrams on impact and in static tests are of a similar character. In this case, the forces required to fracture the limestone under static and dynamic conditions are approximately the same regardless of the rate of loading. R. M. Eigeles [10] likewise found in his work no significant differences between the values of the forces required to fracture rock under static and dynamic loading conditions.

At the same time, other studies [4, 5, 11–13] have noted an increase in the fracture resistance of rock with increasing penetration rate. For example, when a wedge was pressed into a limestone block, "force–penetration" curves were obtained that differed for static conditions and for impact action performed at different velocities [14]. As can be concluded from the graph in Fig. 17 of [14], the forces corresponding to the moment of rock fracture under dynamic conditions are from 3 (for an impact velocity of 3.1 m/s) to 6 (5.4 m/s) times higher than the fracture force under static loading. Other information concerning rock fracture by impact and under quasi-static conditions is given in publications [15–23].

In the present work, the above-mentioned discrepancies in the assessment of the forces required to fracture rock under static and dynamic conditions are verified by an experimental and computational approach.

Experimental procedure

The object of fracture was sandstone from the Osinovsky massif with a hardness of f = 4 on the M. M. Protodyakonov scale and a compressive strength of $\sigma_{сж}$ = 82–112 MPa, in the form of 40×40 mm plates with a thickness of δ = 10, 15 and 30 mm cut from monolithic pieces along the bedding. The penetrating tool — a cone-shaped indenter with a hemispherical blunting area of radius R = 3 mm, fixed on a cylindrical rod of length L = 175 mm and mass M = 2 kg — was made of U8 tool steel with a hardness of ~60 HRC.

Experiments under quasi-static conditions were carried out using a Zwick ZHU 187.5 testing machine and consisted in comparatively slow ($1.3 \cdot 10^{-5}$ m/s) penetration of the indenter into the midpoint of the plate surface (Fig. 1) until the plates split, with recording of the dependences of force N on the displacement h of the machine crossheads, from whose form the following were determined:

- the maximum penetration of the indenter into the rock α_p preceding splitting of the plate;
- the splitting force N_p;
- the stiffness coefficient $g_{ст}$ of the indenter–rock system;
- the splitting work A_p;
- the specific energy consumption of splitting $A_{p.уд} = A_p/F_p$, where F_p is the area of the surface formed by the split fragments;
- the contact fracture stress $\sigma_{к.р} = N/F_к$, where $F_к$ is the area of the contact surface of the indenter with the rock.

[Figure 1 — see original PDF, page 5] Zwick ZHU 187.5 machine prepared for testing (a) and view of the tool at the moment of indenter penetration into a sandstone plate (b)

Experiments under dynamic conditions were carried out using an experimental test bench [24] that allows the same tool to fall freely inside a tubular holder from a specified height H, with the indenter striking the plate. The impact energy (Q_{max} = 10 J) and impact velocity (v_{max} = 2.2 m/s) were calculated from the mass of the falling tool and its drop height. In the same experiments, oscillograms of the impact loads were recorded by placing under the plate a PK 1000 piezoelectric force sensor connected to an NI USB 6008 analog-to-digital converter; from their form, the time t_m to reach the maximum force on impact was determined.

In the first series of dynamic tests, using a dial indicator with a measurement error of 0.01 mm, the relationship was established between the depth $\alpha_л$ of the spall craters formed on the plate as a result of its contact fracture on impact and the impact energy Q. For each energy value, 10 impacts were applied to different points of the plate (δ = 30 mm), spaced at least three crater diameters apart, and from the results the arithmetic mean of $\alpha_л$ and its standard deviation σ_α were calculated.

In the second series, the minimum energy Q_p required to split plates 10 and 15 mm thick by an impact at the midpoint of their surface, and the specific energy consumption of the splitting process $Q_{p.уд} = Q_p / F_p$, were determined.

Results of experiments under quasi-static conditions

Fig. 2 illustrates the typical form of the N–h dependences obtained; the end point of the curve with coordinates α_p, N_p corresponds to the moment of plate splitting.

[Figure 2 — see original PDF, page 6] Typical curve of force versus displacement of the machine crossheads during indenter penetration into a sandstone plate

At the initial stage I, displacement occurs without a noticeable increase in load; at stage II the load increases according to a linear dependence close to that described in the literature. Since at displacement stage I there are no traces of indenter penetration into the rock, it can be concluded that this stage corresponds to taking up the clearances between the moving elements of the machine fixture; consequently, only stage II of the diagram corresponds to indenter penetration, with

the displacement expressed by the quantity α on the graph. By extrapolating the dependence N = f(α) with a straight line, the angle β of its inclination to the displacement axis was determined, whose tangent was taken as the stiffness $g_{ст}$ of the indenter–plate system under quasi-static conditions, with the conclusion that the load characteristic of contact fracture of sandstone during indenter penetration is described by a force function of the form $N = g_{ст}\alpha$.

The contact area $F_к$ was taken as the projected area of the surface of the embedded part of the indenter, calculated by the formula $F_к = 2\pi\alpha(2R - \alpha)$ [25]. Since, owing to R (3 mm) being considerably larger than α (up to 0.37 mm), the dependence $F_к = f(\alpha)$ proved to be close to linear, the stress $\sigma_{к.р} = N/F_к$ at which contact fracture of sandstone occurs was assumed constant over the whole range of α.

From the mean values of the established fracture parameters $\alpha_р$, $N_р$, the plate splitting work was calculated as $A_р = (\alpha_р N_р) / 2$. Taking into account the appearance of the fractured plates, $F_р$ was taken as the product of the length of the split and the plate thickness.

The values of the established parameters of the splitting process and contact fracture of the sandstone plates are given in Table 1.

Table 1. Parameters of splitting and contact fracture of plates during indenter penetration into sandstone under quasi-static conditions

Test No.	$\alpha_р$, mm	$N_р$, N	$A_р$, J	$A_{р.уд}$, kJ/m^2	$g_{ст}\cdot10^{-7}$, N/m	$\sigma_{к.р}$, MPa
δ = 10 mm						
1	0.21	2127	0.22	0.56	1.01	470
2	0.20	3043	0.30	0.76	1.52	400
3	0.18	2330	0.20	0.51	1.33	384
4	0.32	3299	0.53	1.32	1.03	440
5	0.32	2909	0.47	1.16	0.91	480
6	0.23	2498	0.28	0.70	1.11	547
Mean value	0.24	2734	0.33	0.84	1.15	470
δ = 15 mm						
1	0.31	3052	0.47	0.79	0.98	532
2	0.26	3539	0.47	0.78	1.35	427
3	0.32	3044	0.30	0.60	1.05	400
4	0.29	3245	0.47	0.78	1.12	369
5	0.37	4240	0.78	1.30	1.15	399
Mean value	0.30	3224	0.49	0.82	1.13	411

As can be seen from the data in the table, with increasing plate thickness the parameters $\alpha_р$, $N_р$, $A_р$ increase, while $A_{р.уд}$, $g_{ст}$, $\sigma_{к.р}$ do not change appreciably. At the same time, the obtained value of $A_{р.уд}$ (~830 J/m^2) considerably exceeds the value of this parameter (60–90 J/m^2) established in [1] for splitting sandstone plates with wedges, which can be explained by the difference in the methods of rock fracture. Whereas in [1] the action is applied to the fragment along its entire length, in the present work it is applied only through point contact.

It may also be noted that the established value of the contact fracture stress ($\sigma_{k.p}$ ~ 450 MPa) is 4–5 times higher than the compressive strength of the sandstone used, which is explained by the influence of the scale factor: when an indenter penetrates rock, only micro-volumes of the rock are fractured, whereas in compression tests the whole volume of the test specimen, with all the structural imperfections present in it, is involved.

Results of experiments under dynamic conditions

The results of tests on contact fracture of sandstone plates ($\delta_{k.p}$ = 30 mm) on impact are presented in Table 2. They show that, with increasing impact energy Q, the depth $\alpha_л$ of the spall craters in the rock increases.

Table 2. Arithmetic mean $\alpha_л$ and standard deviation σ_α of the spall crater depth in a sandstone plate after a series of impact penetrations of the indenter as a function of impact energy

Q, J	0.4	1.0	2.6	3.3	3.7	4.1	4.6
$\alpha_л \pm \sigma_\alpha$, mm	0.18 ± 0.040	0.26 ± 0.06	0.38 ± 0.07	0.42 ± 0.06	0.45 ± 0.11	0.50 ± 0.06	0.53 ± 0.14

A typical form of the oscillograms obtained is shown in Fig. 3. It turned out that over the whole range of impact velocities v (0.63–2.20 m/s) and energies Q (0.4–4.6 J), the time t_m to reach the maximum load does not go beyond 0.89–0.95 ms, with an average value of ~0.90 ms, which is consistent with the data of [9], where the time to reach the maximum load likewise does not change significantly with changes in impact velocity.

As the experiments of the second test series showed, splitting of sandstone plates 10 mm thick occurs starting from an impact energy of Q_p ~ 4.6 J, and of plates 15 mm thick from 5.0 J. The form of the oscillograms proved similar to those obtained for contact fracture (Fig. 3), with t_m* varying in the range 0.90–1.00 ms and an average value of ~0.96 ms.

[Figure 3 — see original PDF, page 7] Typical oscillogram of the forces arising during impact contact fracture and splitting of sandstone plates (a) and its enlarged view (b)

The values of the other parameters of the plate splitting process on impact, needed for comparison with the results of tests under quasi-static conditions, were estimated by calculation.

Calculation of plate splitting parameters on impact

Considering that the impact tool in the experiments was an elastic element of relatively short length, it was assumed that the influence of wave processes on its impact parameters can be neglected and that the process of interaction between the tool and the rock can be described by the equations of V. B. Sokolinsky's two-element classical impact system [25], with the load characteristic expressed as the linear function N = $g_1\alpha$ established under quasi-static conditions:

$$\alpha_m = \sqrt{(2Q^* / g_1)}, \quad (1)$$

$$N_m = \sqrt{(2g_1 Q^*)}, \quad (2)$$

$$t_m = \pi / (2\sqrt{(m / g_1)}), \quad (3)$$

where α_m is the maximum penetration depth of the tool indenter into the rock, N_m is the maximum impact force, t_m is the time to reach the maximum force in contact fracture and the impact duration in

plate splitting, m is the mass of the impact tool, g_1 is the contact stiffness of the impact system, and Q* is the impact energy expended on rock fracture.

It was assumed that, for calculating the impact parameters by (1)–(3), the mean value of $g_{ст}$ from Table 1 may be taken as the system stiffness coefficient g_1. Since on impact only part of the energy of the impact tool is spent on rock fracture (~54 % for sandstone [26]), while the rest goes to heat losses to the environment, elastic deformation of the tool and wear of its contact surface, the energy expended on rock fracture was taken as Q* = 0.54Q.

The validity of these assumptions was checked by comparing the values of α_m, t_m calculated by formulas (1) and (3) for various impact energies Q* with the crater depth $\alpha_л$ and the value of t_m recorded experimentally at the same energies Q* = 0.54Q for contact fracture of the rock. The calculated form of dependence (1), plotted in $\alpha_m - (Q^*)^{0.5}$ coordinates at $g_1 = g_{ст}$ (curve 1), together with the experimentally determined crater depths $\alpha_л$ and their approximation by a linear function (curve 2), is shown in Fig. 4.

[Figure 4 — see original PDF, page 7] Dependences of α_m (1) and $\alpha_л$ (2) on the impact energy Q* for contact fracture of sandstone plates

As can be seen from Fig. 4, all the experimental points obtained are satisfactorily approximated by dependence (2) $\alpha_л = k_2 Q^{0.5}$. The value of the proportionality coefficient of this dependence (k_2 = $0.32 \cdot 10^{-3}$ m/J) is very close to the value of the analogous coefficient k_1 ($0.43 \cdot 10^{-3}$ m/J) calculated by formula (2) at $g_1 = g_{ст}$. On the other hand, the time to reach the maximum force calculated by formula (3) (t_m = 0.67 ms) for contact fracture of the rock proved close to the experimentally established value (0.90 ms). In the authors' opinion, these circumstances confirm the possibility of calculating the impact parameters in the experiments by expressing the load characteristic as the function $N = g_1\alpha$ using the system stiffness coefficient for quasi-static conditions ($g_1 = g_{ст}$).

Discussion of results

Table 3 gives the values of A_p, $A_{p.уд}$, α_p, N_p corresponding to plate splitting under quasi-static conditions, in comparison with the experimentally established parameters of plate splitting on impact Q_p and $Q_{p.уд}$, t_m*, as well as the impact parameters α_m, N_m, t_m calculated from Q_p* (formulas (1)–(3)).

Table 3. Parameters of splitting of sandstone plates under quasi-static conditions and under impact action (v ~ 2.2 m/s)

Splitting conditions	δ, mm	A_p (Q_p), J	$A_{p.уд}$, kJ/m^2	α_p (α_m), mm	N_p (N_m), kN	t_m* (t_m), ms
Quasi-static	10	0.33	0.55	0.24	3.2	–
	15	0.49	0.6	0.30	4.3	
Dynamic	10	4.6	11.5	0.67	7.4	0.96 (0.67)
	15	5.0	13.0	0.70	7.7	

As a comparison of the parameters in the table shows, the energy Q_p and the specific energy consumption $Q_{p.уд}$ of splitting sandstone plates on impact are more than 10 times higher than these parameters (A_p, $A_{p.уд}$) under quasi-static conditions. This result is consistent with the literature data

and is explained both by energy losses on impact and by the fact that, on passing to dynamic action, the resistance of rocks to plastic deformation and fracture increases.

At the same time, the comparative results obtained for the applied fracture forces — on impact they are approximately twice as high as under static conditions — agree with the data in [4, 14]. As the authors of [14] conclude, the absence of significant changes in force values under static and dynamic loading at low indentation rates established in [10] is explained by the small absolute values of these changes, which are masked by the scatter of force values within a single test rate. Moreover, if it is assumed that the force required to fracture rock is the same for static and dynamic conditions, it becomes unclear why the work of fracture differs so substantially (by a factor of 10) for these conditions. If, however, one accepts the existence of a difference in the magnitude of these forces, as well as the difference found in this work in the indenter penetration depths required to fracture the plate — for dynamic conditions they are approximately twice as large as for static conditions — then the above difference in the values of the work (energy) of fracture becomes explicable and can be calculated from the values of these parameters.

Thus, the results of the study show that, from the energy point of view, rock fracture under comparatively slow action of a rock-breaking tool is preferable to impact action. At the same time, the advantage of using impact systems is the possibility of obtaining forces at the output of the system hundreds and thousands of times greater than those applied at its input [25], which ensures efficient rock fracture by impact machines. In this connection, it can be concluded that, when breaking rock pieces of comparatively small size and low hardness, when large forces are not required, it is advisable to use devices operating under quasi-static conditions, for example jaw crushers. At the same time, for large pieces of hard rock (oversize boulders), whose fracture requires the application of considerable forces, it is preferable to use impact devices, for example hydraulic breakers.

Conclusions

1. Experiments on the penetration of an indenter with a hemispherical blunting area into sandstone plates 10 and 14 mm thick under quasi-static (v ~ 1 mm/min) and dynamic (v_{max} ~ 2.2 m/s) conditions have shown that the work and specific energy consumption of the splitting process under comparatively slow action of a rock-breaking tool on the rock are more than an order of magnitude lower than under impact conditions.
2. Assuming that the load characteristic of contact fracture of a sandstone plate on impact, as under static conditions, is described by a function of the form N = gα with similar system stiffness coefficients, and that only about half (54 %) of the impact energy is spent on splitting the rock, the parameters of indenter penetration into sandstone plates — the maximum penetration depth α_m, the maximum impact force N_m and the impact duration t_m — have been calculated for various impact energies for surface contact fracture of the plates and for their splitting.
3. The reliability of the calculated parameters of contact fracture of the rock is confirmed by the closeness of the calculated penetration depths α_m to the experimentally recorded spall crater depths $\alpha_л$, and by the insignificant difference between the time to reach the maximum force on impact established by calculation and that established experimentally from the form of the oscillograms.
4. As shown by the combined computational and experimental approach, the forces required to split sandstone plates on impact are approximately twice as high as the forces under static

loading conditions, which confirms the literature data in which these parameters differ substantially from each other.

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